IEEE VTS Motor Vehicles Challenge 2021 - Energy Management of A Dual-Motor All-Wheel Drive Electric Vehicle

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***Abstract*—This paper proposes a challenge on the development of energy management strategy (EMS) for a dual-motor all-wheel drive (AWD) electric vehicle (EV). This challenge follows the success of the previous four IEEE VTS Motor Vehicles Challenge 2017–2020. The challenge welcomes participants from both academic and industrial sectors. An urban EV available at the University of Sherbrooke is used as a reference. The AWD traction configuration is considered as a combination of two electrical machines installed in the front and rear axles of the vehicle. Their associated inverters are supplied by a Lithium-ion battery. The objective of the challenge is the EMS that leads to the minimal state-of-charge of the battery for a given driving cycle. A downloadable MATLAB/Simulink model and control of the vehicle organized by using Energetic Macroscopic Representation (EMR) are provided. The top-scored participated teams will be distinguished and invited to present their EMSs in a special session of the IEEE-VPPC 2021.**

***Keywords*—Energy management strategy, Li-ion battery, Energetic Macroscopic Representation (EMR).**

I. INTRODUCTION

Initiated in the IEEE Vehicle Power and Propulsion Conference (VPPC) 2016 in Hangzhou, China [1], the IEEE VTS Motor Vehicles Challenge has been successfully organized over four versions from 2017 to 2020. The challenges focused on development of energy management strategy (EMS) for a fuel-cell/battery vehicle [2], [3], a range extender vehicle [4], [5], a dual-mode locomotive [6], [7], and a fuel-cell/battery/supercapacitor (SC) hybrid electric vehicle (HEV) [8]. In these previous challenges, the studied systems are HEVs and hybrid energy storage systems.

Meanwhile, dual-motor all-wheel drive (AWD) is an advanced traction configuration being developed in both academic and industrial sectors [9]–[13]. In industry, for example, Tesla Model 3 is equipped with an interior permanent magnet motor in the rear and an induction motor in the front axle [14]. For academic research, a front- and rear-wheel independent drive (FRID) EV prototype has been developed with high performances of dynamical behaviors and fault-tolerance ability [15]–[17]. Recently, the impacts of different efficiency characteristics of the front and rear electrical drives (EDs) have been investigated [18]. Nevertheless, the number of works on EMS of this configuration is still limited.

This paper proposes a challenge on the design of an EMS for a dual-motor AWD EV of which the reference model is given from an urban EV developed in the University of Sherbrooke (Fig. 1). Two EDs with different efficiency maps are installed in the front and rear axles of the vehicle. The participants will develop their EMS to allocate the traction forces between the front and rear axles and also the mechanical and electrical braking forces of each axel. The ultimate objective of the EMS is to minimize the battery SoC depletion respecting the torque constraints of the EDs and the temperature constraints of the battery cell.

Energetic Macroscopic Representation (EMR) is a graphical formalism utilized for model organization for control purpose [19]. EMR has been applied for modeling, control, and energy management of complex systems including dual-motor single-inverter drive [20], transmission with clutch in HEVs [21], fuel-cell [22], fuel-cell/battery/SC heavy-duty vehicle [23], diesel-electric locomotive [24], and battery/SC EV [25]. In this challenge, EMR is used as a common formalism for modeling and control of the studied system.



Fig. 1. Urban electric vehicle VUE available at the University of Sherbrooke.

In this paper, the configuration of the studied vehicle and the modeling and control organized by using EMR are addressed in Section II. The EMS framework, the scoring, and the MATLAB/Simulink file are then mentioned in Section III. Finally, conclusion and outlook are given in Section IV.

II. Modeling and Control of The Studied Vehicle

The configuration of the dual-motor AWD EV under study is described in Fig. 2. Two different three-phase AC electrical machines (EMs) are installed in the front and rear axles of an EV. The machines are driven by using the associated inverters of which the DC buses are connected in parallel with a Lithium-ion battery. The battery EMR description is given in Fig. 3 which is an electro-thermal model adapted from [26]. In this challenge, the electrical drives combining the EMs and the inverters are considered by using the efficiency maps given in Fig. 4. The vehicle model organization using EMR is given in Fig. 5 with the pictograms explained in Appendix. The mathematical models implemented inside the EMR blocks are as follows.

The battery cell electrical model is given by:

$$\begin{cases} u_{\text{cell}} = u_{\text{cell OC}}(SoC_{\text{cell}}, T_{\text{cell}}) - r_{\text{cell}}(SoC_{\text{cell}}, T_{\text{cell}})i_{\text{cell}}; \\ SoC_{\text{cell}} = SoC_{\text{cell}}(0) - \dfrac{1}{C_{\text{eq}}}\displaystyle\int_0^t i_{\text{cell}}dt\,; \end{cases} \quad (1)$$

in which the nonlinear relationships of the cell open-circuit voltage $u_{\text{cell OC}}$ and the equivalent series resistance (ESR) r_{cell} with respect to the SoC_{cell} and T_{cell} are given as lookup tables. Due to the series and parallel connections of the cells, the relationships between battery and cell voltages and currents can be given by:

$$\begin{cases} u_{\text{bat}} = u_{\text{cell}} n_{\text{se}}; \\ i_{\text{cell}} = \dfrac{i_{\text{bat}}}{n_{\text{pa}}}; \end{cases} \quad (2)$$

where n_{se} is the number of cells connected in series and n_{pa} is the number of parallel branches of serial cells.

To couple the thermal and the electrical models of a battery cell, the cell entropy flow (in W/K) should be introduced as:

$$q_{\text{cell}} = \frac{r_{\text{cell}}(SoC_{\text{cell}}, T_{\text{cell}})i_{\text{cell}}^2}{T_{\text{cell}}}; \quad (3)$$

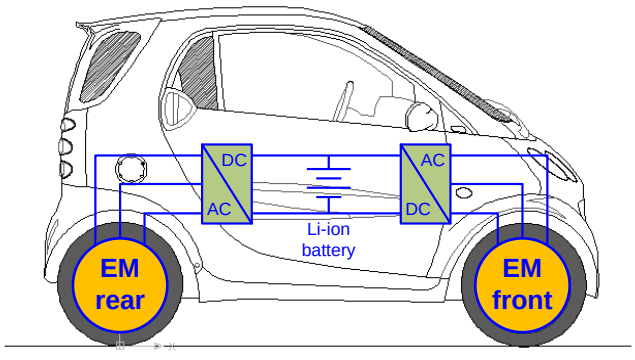


Fig. 2. Studied dual-motor AWD configuration.

in which T_{cell} is the cell core temperature given by:

$$T_{\text{cell}} = T_{\text{cell}}(0)e^{\frac{1}{C_{\text{th}}}\int_0^t (q_{\text{cell}} - q_{\text{cs}})dt} \quad (4)$$

where C_{th} is the equivalent thermal capacitance; q_{cs} the entropy flow of the heat conduction from the core to the surface of the cell. Similarly, the entropy flow of the heat conduction from the cell surface to the air is denoted by q_{sa}. These entropy flows are calculated by:

$$\begin{cases} q_{\text{cs}} = \dfrac{T_{\text{cell}} - T_{\text{air}}}{r_{\text{th}} T_{\text{cell}}}; \\ q_{\text{sa}} = \dfrac{T_{\text{cell}} - T_{\text{air}}}{r_{\text{th}} T_{\text{air}}}; \end{cases} \quad (5)$$

with r_{th} is the equivalent thermal resistance.

The electrical drives are considered by using the static models, in which the drive torque Γ_{EDx} immediately responds to the reference, as follows:

$$\begin{cases} \Gamma_{\text{EDx}} = \Gamma_{\text{EDx ref}} \\ i_{\text{EDx}} = \dfrac{\Gamma_{\text{EDx}}\Omega_{\text{EDx}}}{u_{\text{bat}}\eta_{\text{EDx}}^{k_{\text{EDx}}}} \text{ with } k_{\text{EDx}} = \begin{cases} 1 \text{ if } \Gamma_{\text{EDx}}\Omega_{\text{EDx}} \geq 0 \\ -1 \text{ if } \Gamma_{\text{EDx}}\Omega_{\text{EDx}} < 0 \end{cases} \end{cases} \quad (6)$$

where "x" replaced by "f" and "r" denoting the front and rear axles, respectively. It should be noted that the effect of the efficiency η_{EDx} changes regarding the power direction reflected by k_{EDx}. Such a static model is sufficient for study of energy consumption [27].

The gearboxes, the differentials, and the wheels associated with the front and rear axles are combined as the transmission blocks given by:

$$\begin{cases} F_{\text{Tx}} = \Gamma_{\text{EDx}}\eta_{\text{tran}}^{k_{\text{trx}}} k_{\text{tran x}} \\ \Omega_{\text{EDx}} = v_{\text{veh}} k_{\text{tran x}} \end{cases} \text{ with } k_{\text{trx}} = \begin{cases} 1 \text{ if } F_{\text{Tx}} v_{\text{veh}} \geq 0 \\ -1 \text{ if } F_{\text{Tx}} v_{\text{veh}} < 0 \end{cases}; \quad (7)$$

in which the transmission efficiency η_{tran} is a constant.

The mechanical brakes impose the braking force F_{brx} to the wheel of the vehicle by following the reference $F_{\text{brx ref}}$ as:

$$F_{\text{br}x} = F_{\text{brx ref}}. \quad (8)$$

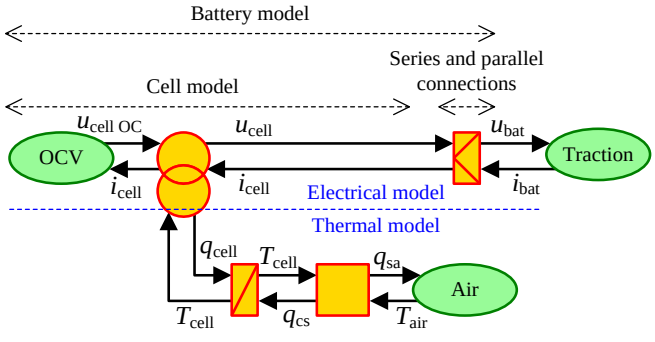
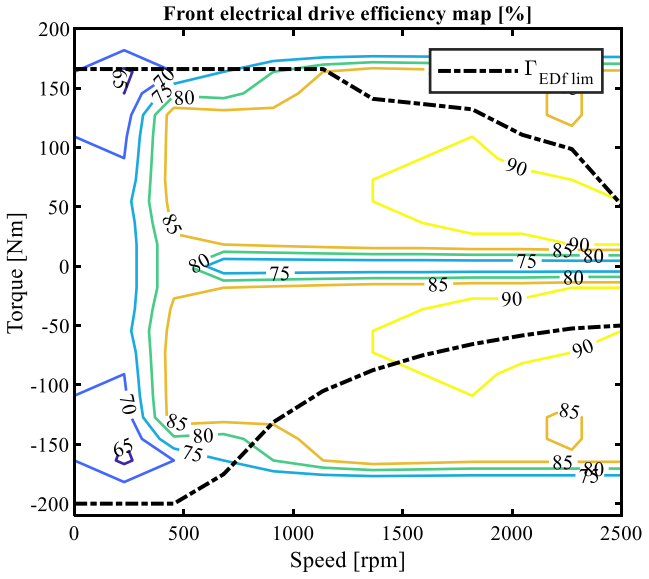
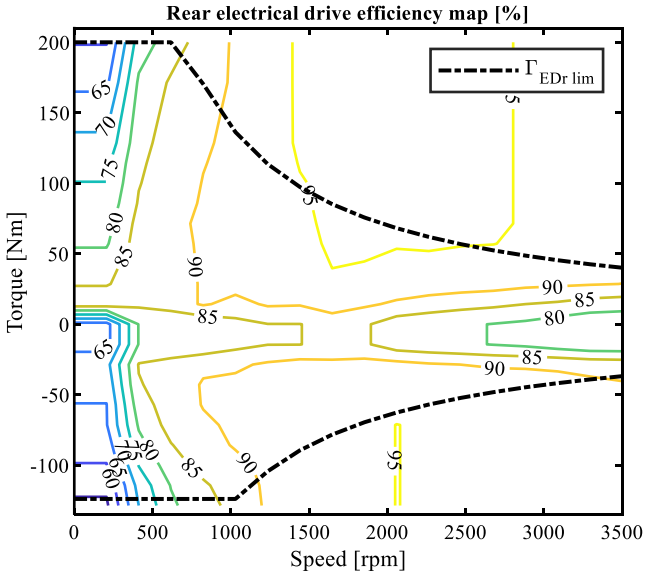


Fig. 3. Battery electro-thermal model.



(a)



(b)

Fig. 4. Front and rear electrical drives efficiency maps.

with $F_{\text{brx ref}} \leq 0$. Here, it is assumed that there is no dynamical delay between the reference and the response of the braking forces (ideal brake control).

The brake couplings blocks link the transmission forces F_{Tx} and the mechanical braking forces F_{brx} as follows:

$$\begin{cases} F_{\text{x}} = F_{\text{Tx}} + F_{\text{brx}}; \\ v_{\text{veh}} \text{ common}; \end{cases} \tag{9}$$

in which F_{x} is the front and rear traction forces regarding the replacement of "x" by "f" and "r", respectively.

The axles traction forces F_{f} and F_{r} are linked at the chassis coupling to impose the total traction force F_{trac} as given by:

$$\begin{cases} F_{\text{trac}} = F_{\text{f}} + F_{\text{r}}; \\ v_{\text{veh}} \text{ common}. \end{cases} \tag{10}$$

The chassis model, the environment, and the vehicle speed controller are similar to which have already been given in [25]. It includes a velocity controller to represent the driver and deliver the total reference force $F_{\text{trac ref}}$ to move the car.

The axles traction forces allocation is realized by introducing a traction distribution factor k_{fr} as follows:

$$\begin{cases} F_{\text{f ref}} = k_{\text{fr}} F_{\text{trac ref}}; \\ F_{\text{r ref}} = (1 - k_{\text{fr}}) F_{\text{trac ref}}. \end{cases} \tag{11}$$

The braking forces allocations are similarly implemented by using the braking distribution factors k_{brx} given by:

$$\begin{cases} F_{\text{Tx ref}} = k_{\text{brx}} F_{\text{x ref}}; \\ F_{\text{brx ref}} = (1 - k_{\text{brx}}) F_{\text{x ref}}; \end{cases} \tag{12}$$

with the replacement of "x" by "f" and "r" for the front and rear axles, respectively.

The inversions of the transmissions are also similar to the one realized in [25].

III. Strategy Framework and Scoring

The participants have to develop a real-time EMS which deduce the distribution factors k_{fr}, k_{brf}, and k_{brr} for a driving cycle which is not *a priori* known. These factors decide the allocations of the front and rear axles traction forces and of the braking forces of the corresponding wheels, respectively. Regarding the different allocation strategies, the distributions of the operation points in the efficiency maps will be different. Each EMS will therefore conduct particular losses on the EDs and the battery due to its ESR. As a result, a certain SoC evolution will be deduced for each strategy.

The objective of the EMS is to minimize the battery SoC depletion. The final SoC of the battery is the merit to score the effectiveness of the examined strategy. The EDs torques must be constrained by their limitations (see Fig. 4). Furthermore, the battery cells should not be neither too hot nor too cold to avoid the harmfulness to their lifespan. Hence, the energy management problem can be formulated as follows:

$$\begin{aligned} [k_{\text{fr}} \quad k_{\text{brf}} \quad k_{\text{brr}}] &= \arg\min [SoC_{\text{bat}}(0) - SoC_{\text{bat}}(t_{\text{f}})] \\ \text{s. t.} \quad & \Gamma_{\text{EDf min}} \leq \Gamma_{\text{EDf}} \leq \Gamma_{\text{EDf max}} \\ & \Gamma_{\text{EDr min}} \leq \Gamma_{\text{EDr}} \leq \Gamma_{\text{EDr max}} \\ & T_{\text{cell min}} \leq T_{\text{cell}} \leq T_{\text{cell max}}; \end{aligned} \tag{13}$$

and subject to the system model and control addressed in Section II as the equality constraints; in which t_{f} is the final time of the driving cycle. It should be noted that in some critical case, e.g., starting the vehicle after battery fast charging, the cell may be at high initial temperature. Hence, after a long driving cycle with high traction and regenerative power exchanges, the cell temperature may reach its limitation. That constraint must be considered by the EMS.

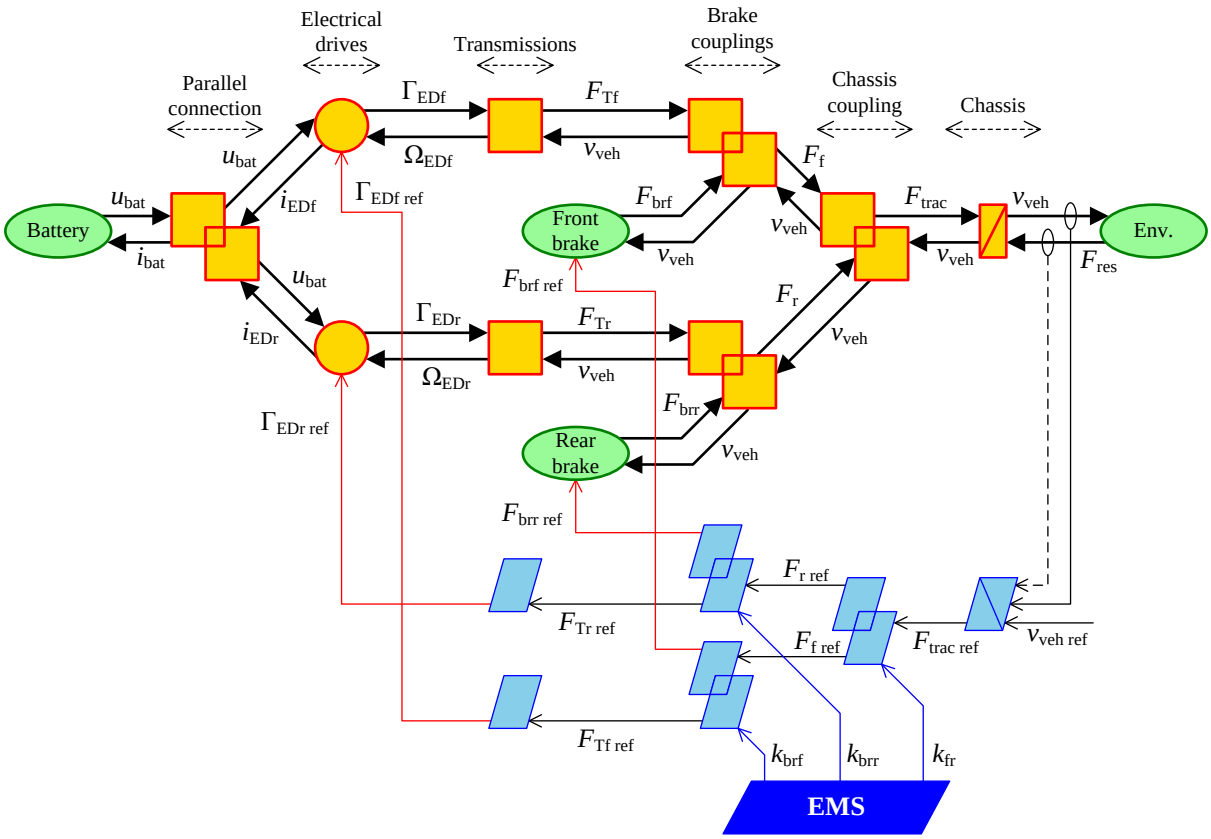


Fig. 5. EMR and inversion-based control scheme of the studied vehicle.

In order to focus on the EMS design and for a fair evaluation, the developed strategies will be compared by using a common simulation of the model and control scheme. For that, the MATLAB/Simulink-based simulation files of the studied system are openly provided as illustrated in Fig. 6. The participants will implement their strategy in the EMS block (dark blue background with the white parallelogram) by using the provided measurements. The participants must not change anything of the system model and control. The EMSs will be scored by using the original files of the organization team.

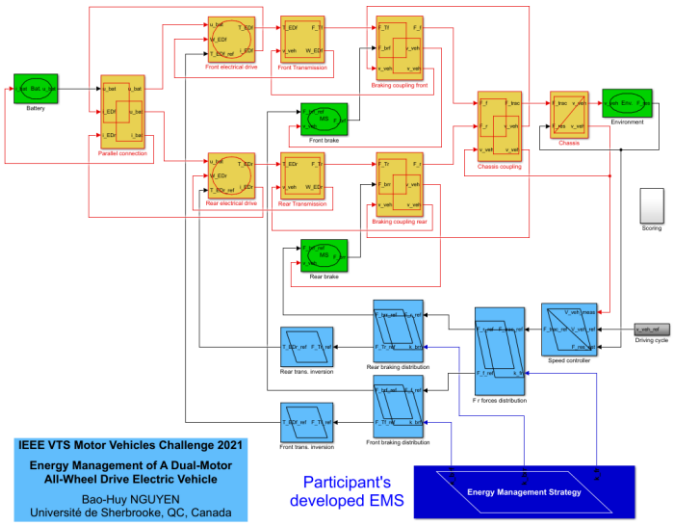
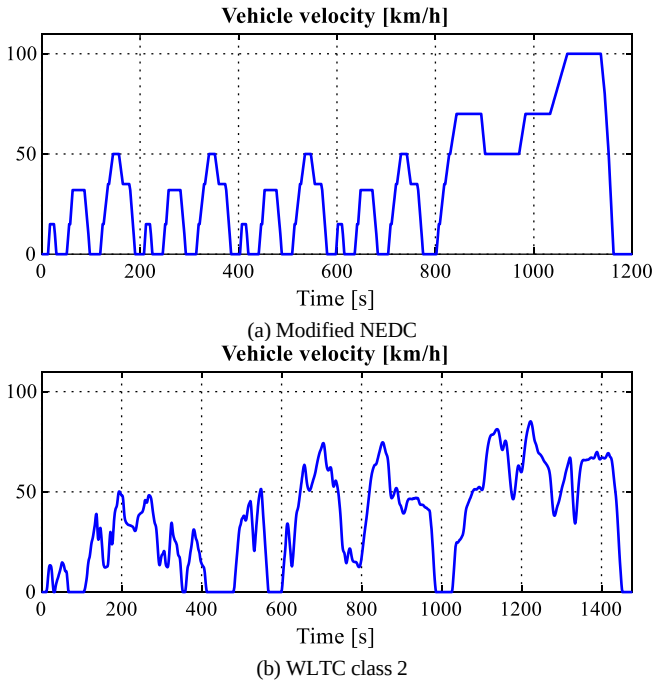


Fig. 6. Simulation in MATLAB/Simulink.

Two driving cycles, which are the modified New European Driving Cycle (NEDC) and the Worldwide harmonized Light vehicles Test Cycles (WLTC) class 2, are provided (Fig. 7) as the samples of the testing condition. However, the examined EMSs will be scored by using a third unknown driving cycle.



(a) Modified NEDC

(b) WLTC class 2

Fig. 7. Given samples of driving cycles.

For better evaluation of the real-time EMSs, the developed strategies will be evaluated not only by comparing together, but also to a global optimal benchmark. Dynamic programming will be applied for obtaining that offline optimal solution.

IV. Conclusion

The paper has proposed a contest for the IEEE VTS Motor Vehicles Challenge 2021. The challenge is to develop a real-time EMS for forces allocations of a dual-motor AWD EV based on the reference model of an available urban EV. The objective of the strategies is to minimize the depletion of the battery SoC by reducing the losses of the EDs and the battery. MATLAB/Simulink-based simulation files of the system model and control have been provided. That will lead to an extension of the driving range of the vehicle. The participants obtaining the best results will be invited to present their works at the IEEE-VPPC 2021. The results will be also compared to a dynamic programming-based optimal benchmark to see how they could be improved.

V. Acknowlegment

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Appendix

Fig. 8 shows the EMR pictograms.

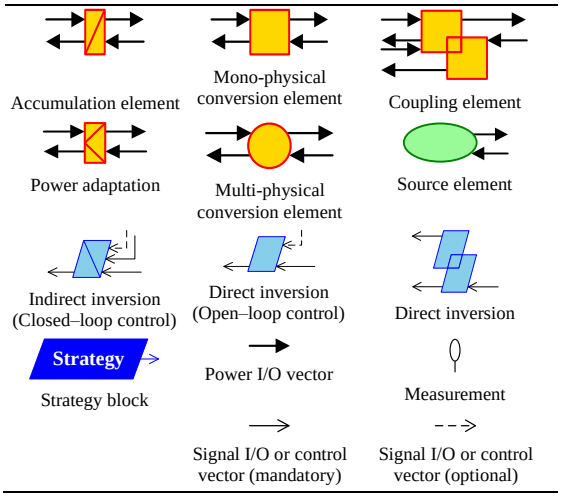


Fig. 8. EMR pictograms.